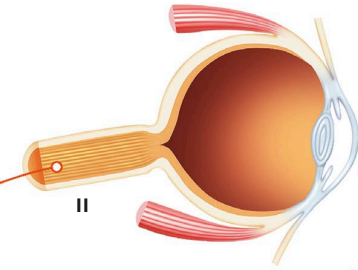


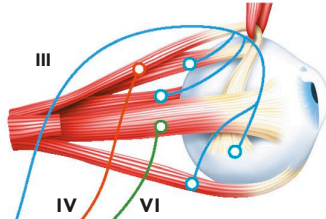
Optic nerve (II, sensory)

The optic nerve sends visual messages from the rod and cone cells in the retina of the eye to the visual cortex in the brain; parts of the two nerves cross at the optic chiasm (see p.122), where they form bands of nerve fibres, called optic tracts. Each nerve consists of a bundle of about one million sensory fibres – it carries the most information of any cranial nerve.



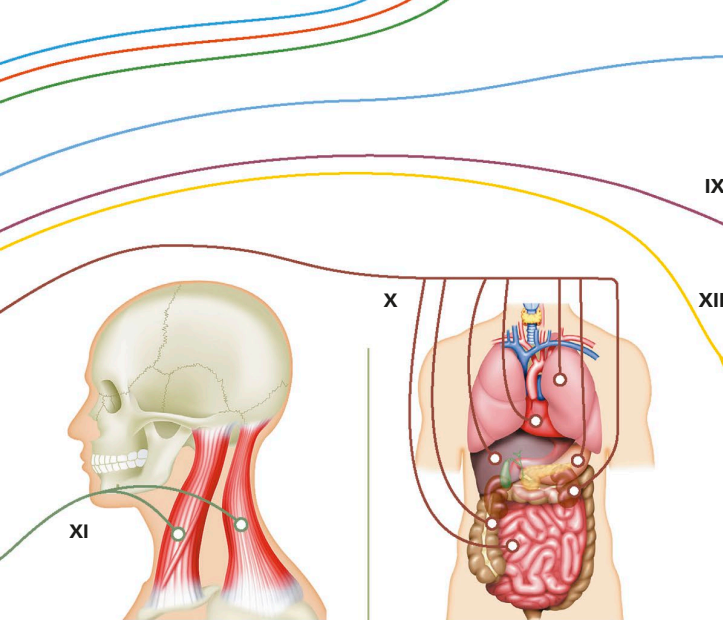
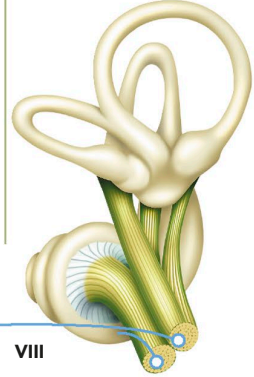
Oculomotor, trochlear, and abducens nerves (III, IV, VI, mainly motor)

These three nerves regulate the voluntary movements of the eye muscles, to move the eyeball and eyelids; the oculomotor also controls pupil constriction by the iris muscles and focusing changes in the lens by the ciliary muscles.



Vestibulocochlear nerve (VIII, sensory)

The vestibular branch sends nerve signals from the inner ear about head orientation and balance; the cochlear branch brings signals from the ear concerning sound and hearing.



Spinal accessory nerve (XI, mainly motor)

This nerve controls muscles and movements in the head, neck, and shoulders. It also stimulates the muscles of the pharynx and larynx, which are involved in swallowing.

Vagus nerve (X, mixed)

The longest and most branched cranial nerve, the vagus has sensory, motor, and autonomic fibres that pass to the lower head, throat, neck, chest, and abdomen; they are involved in many vital bodily functions, including swallowing, breathing, and heartbeat.

Glossopharyngeal and hypoglossal nerves (IX, XII, both mixed)

Motor fibres of these nerves are involved in tongue movement and swallowing, while sensory fibres relay information about taste, touch, and temperature from the tongue and pharynx.

